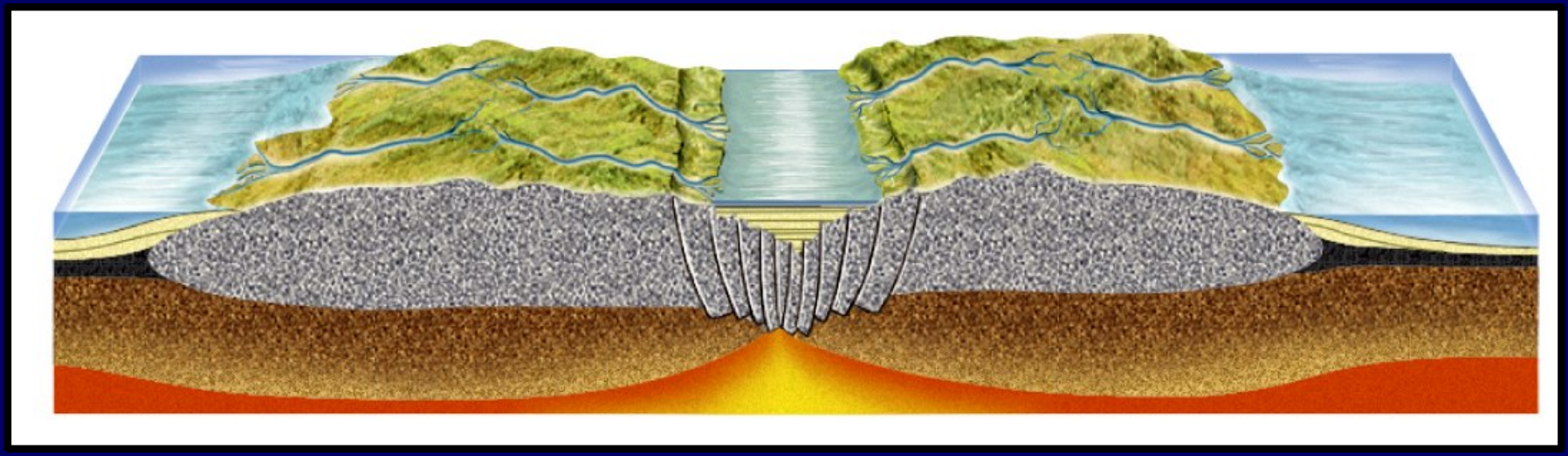
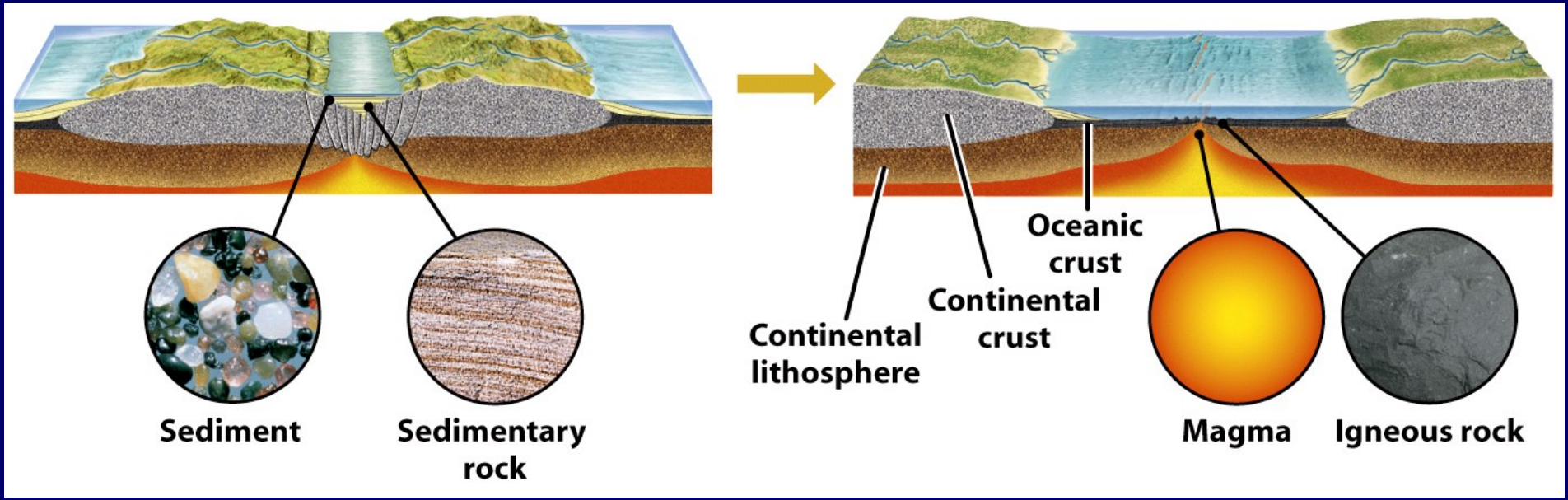
The Rock Cycle



- The three families of rocks (igneous, sedimentary and metamorphic) are related
- The rock cycle explains how each type can be transformed into one of the other two
- The rock cycle is the result of interactions between the plate tectonic and climate systems

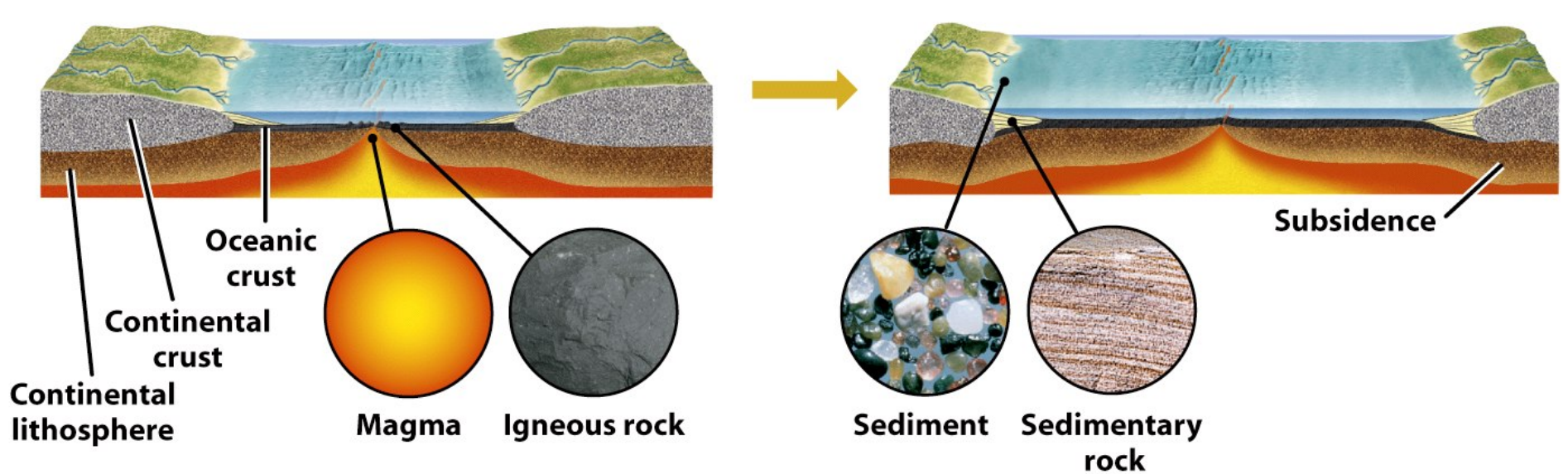
The Rock Cycle



Stage 1
Continental rifting and development of divergent margin
Erosion (chemical and physical weathering) and transportation of sediment from interior by rivers
Deposition of sediment in rift basin
Burial and lithification to form sedimentary rocks

Stage 2
Ongoing seafloor spreading leads to development of new ocean basin
Magma from asthenosphere flows from mid-ocean ridge to form basalt

The Rock Cycle

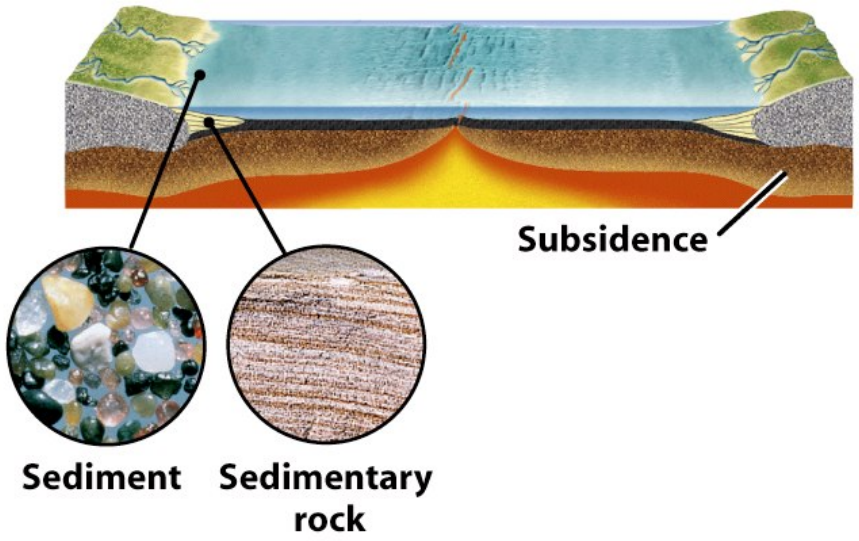


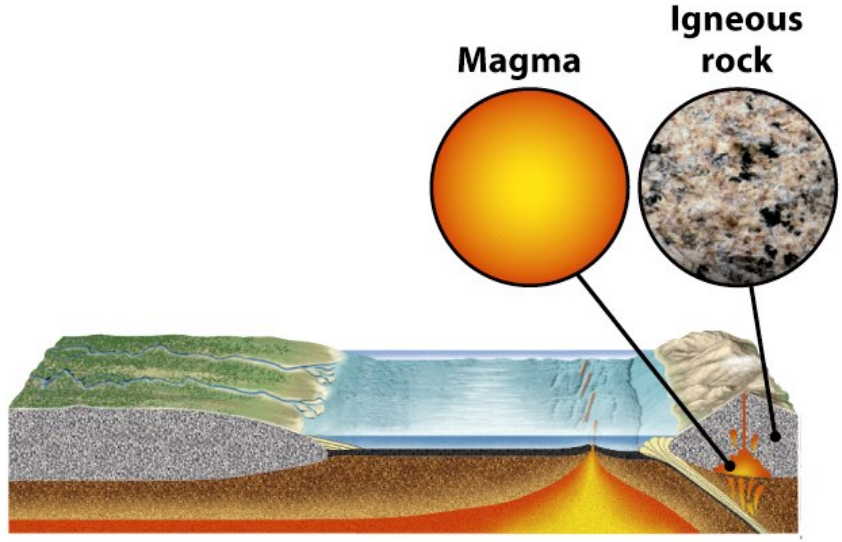
Stage 2
Ongoing seafloor spreading leads to development of new ocean basin
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Stage 3
Subsidence of the continental margin- sinking of the Earth's lithosphere- leads to accumulation of sediment and formation of sedimentary rock during burial

The Rock Cycle

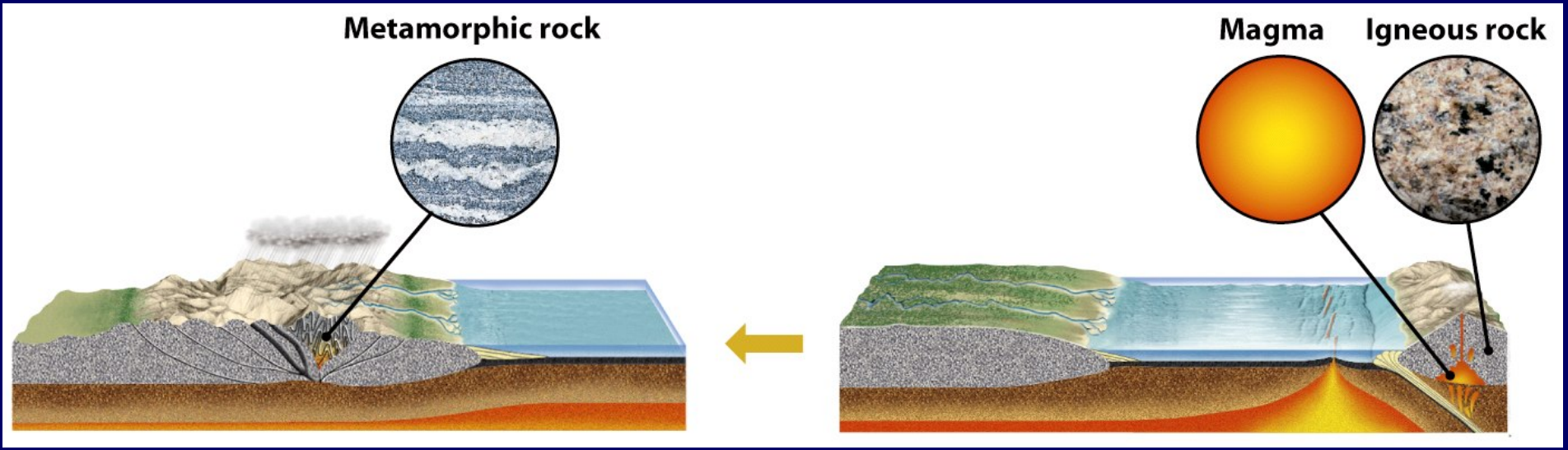
Stage 3
Subsidence of the continental margin- sinking of the Earth's lithosphere- leads to accumulation of sediment and formation of sedimentary rock during burial





Stage 4
Oceanic crust and sedimentary rocks from continental margin subduct beneath a continent. Subducting plate melts as it descends. Magma rises from melting plate and mantle to form volcanic mountain chain and cools to make granitic igneous rocks. The volcanoes inject ash and CO_2 into the atmosphere, affecting global climate

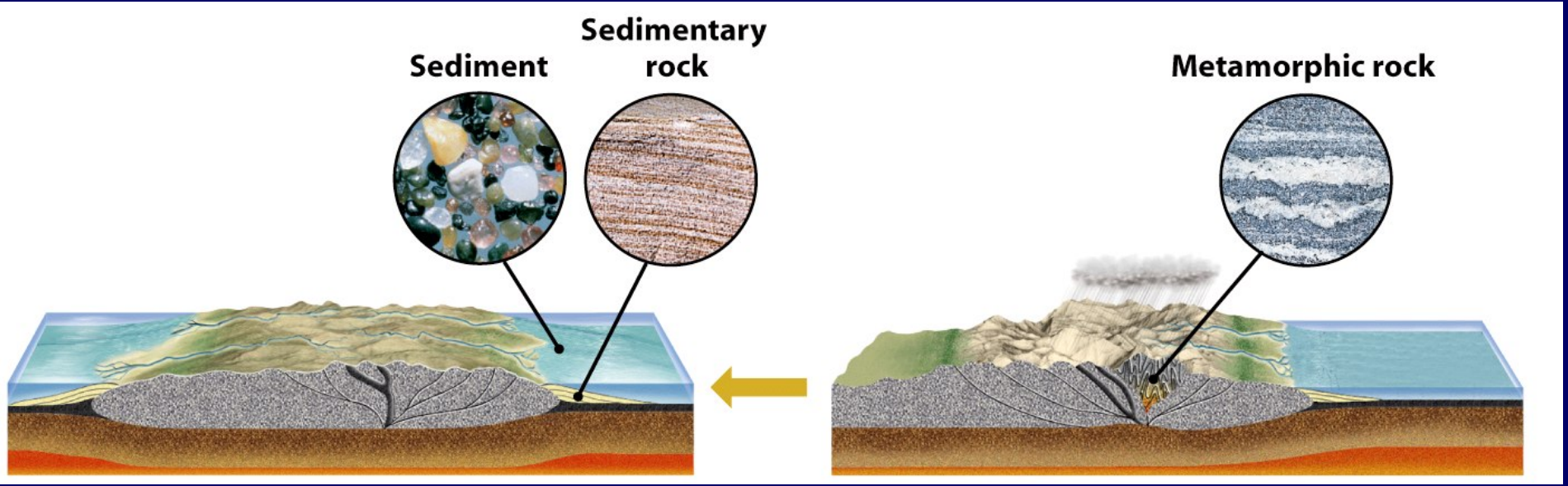
The Rock Cycle



Stage 5
Further closure of the ocean basin leads to continental collision, forming high mountain ranges. Where continents collide, rocks are buried deeper and modified by heat and pressure, forming metamorphic rocks.
Uplifted mountains force moisture-laden air to rise, cool, condense, and precipitate. Weathering creates sediment that can be eroded

Stage 4
Oceanic crust and sedimentary rocks from continental margin subduct beneath a continent. Subducting plate melts as it descends. Magma rises from melting plate and mantle to form volcanic mountain chain and cools to make granitic igneous rocks

The Rock Cycle



Stage 6
Rivers transport sediment away from collision zones to oceans, where it is deposited as layers of sand and silt. Layers of sediment is buried and lithified to form sedimentary rocks

Stage 5
Further closure of the ocean basin leads to continental collision, forming high mountain ranges. Where continents collide, rocks are buried deeper and modified by heat and pressure, forming metamorphic rocks.
Uplifted mountains force moisture-laden air to rise, cool, condense, and precipitate. Weathering creates sediment that can be eroded

The Rock Cycle

Shows only one variant

Long time scales involved e.g., may take billions of years to uplift and expose igneous and metamorphic rocks from deep within crust

Rock cycle never ends

This cycle is at different stages in different places in world

